Le (Leonard) Zhang

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Education

Ph.D. in Computer Science, University of Michigan, Ann Arbor — Expected May 2029
Advisor: Ke Sun

M.S. in Computer Science, University of California, San Diego — June 2025
Advisor: Tajana Šimunić Rosing

B.A. in Computer Science, University of North Carolina at Chapel Hill — December 2022
Major in Computer Science and Minor in Mathematics
Advisor: Shahriar Nirjon

Research Interests

- Resource-efficient AR/VR/wearable vision systems: networking and machine perception.
- Multimodal agents and agentic memory system: efficiency and life-long memory.
- Low-power and efficient multimodal on-device learning.

Awards & Grants

Dean's List: University of North Carolina at Chapel Hill	2019—2022
Honorable Mentioned Award: ICPC Mid-Atlantic Regional	2019

Conference Publications

1. **Le Zhang***, Quanling Zhao*, Run Wang, Shirley Bian, Onat Gungor, Flavio Ponzina, and Tajana Rosing. "Offload Rethinking by Cloud Assistance for Efficient Environmental Sound Recognition on LPWANs." *The ACM Conference on Embedded Networked Sensor Systems (SenSys),* 2025. (* for Equal Contribution, Full Paper, Core Rank A*, Acceptance Rate: 19.2%)
2. **Le Zhang**, Onat Gungor, Flavio Ponzina, and Tajana Rosing. "E-QUARTIC: Energy Efficient Edge Ensemble of Convolutional Neural Networks for Resource-Optimized Learning." *The Asia and South Pacific Design Automation Conference (ASP-DAC),* 2025. (Core Rank B, Acceptance Rate: 31%)
3. Yubo Luo, **Le Zhang**, Zhenyu Wang, and Shahriar Nirjon. "Antler: Exploiting Task Affinity for Efficient Multitask Learning on Low-Resource Systems." *The International Conference on Embedded Wireless Systems and Networks (EWSN),* 2024. (Core Rank B, Acceptance Rate: 22.8%)

Short Papers & Abstracts

1. Run Wang*, Shirley Bian*, Xiaofan Yu, Quanling Zhao, **Le Zhang**, and Tajana Rosing. "Poster: Resource-Efficient Environmental Sound Classification Using Hyperdimensional Computing." *The ACM Conference on Embedded Networked Sensor Systems (SenSys),* 2024. (Core Rank A*)
2. **Le Zhang**, Yubo Luo, and Shahriar Nirjon. "Demo Abstract: Capuchin: A Neural Network Model Generator for 16-bit Microcontrollers." *The ACM/IEEE Conference on Information Processing in Sensor Networks (IPSN),* 2022. (Core Rank A*)

Manuscripts

1. **Le Zhang**, Kailai Cui, Hao Chen, Vlad Roznyatovskiy, Jianzhong Zhang, Anhong Guo, Kang G. Shin, and Ke Sun. "EventHOI: Event-Driven Energy-Efficient Hand-Object Interaction Logging on Smartglasses." Manuscript under review.

Research Experience

GSRA, Ambient Intelligence (AmI) Lab, University of Michigan 2025–2026

EventHOI: Event-Driven Energy-Efficient Hand-Object Interaction Logging on Smartglasses.
Advisor: Prof. Ke Sun

- Designed an event-driven sensing and communication system for energy-efficient hand-object interaction (HOI) logging on smartglasses, leveraging low-power event sensing and IMU-assisted egomotion cancellation to trigger RGB capture only during meaningful interactions.
- Developed lightweight on-device algorithms for HOI detection and saliency-guided video compression, reducing always-on backend power consumption by up to $10.4\times$ and uplink bandwidth by $2.7\times$ while maintaining downstream vision-language model performance.

Student Researcher, System Energy Efficiency Lab (SeeLab), UC San Diego 2024

Resource-Constrained Collaborative Environmental Sound Recognition over LPWANs.
Advisor: Prof. Tajana Rosing

- Proposed a novel cloud-assisted offloading for resource-efficient learning on low-power wide-area networks (LPWANs) like LoRa. Designed for adaptive collaboration for resource-constrained energy-harvesting systems and low-bit rate, unreliable wireless channels.
- Achieved 2.5-12.5% improvement in accuracy, 80x in energy savings, and 220x in latency reduction compared to the state-of-the-art methods.
- 12-page full paper published in *SenSys 2025*.

Student Researcher, the Systems Energy Efficiency Lab (SeeLab), UC San Diego 2023—2024

Energy Efficient Ensemble Learning on Energy-Harvesting Systems.
Advisor: Prof. Tajana Rosing

- Developed an energy-adaptive ensemble learning framework for efficient inference and training on STM32 microcontroller. Improved the reliability of energy-harvesting machine learning system in low-energy conditions up to 40%.
- Paper published in *ASP-DAC 2025*.

Undergraduate Research Assistant, Embedded Intelligence Lab, UNC-Chapel Hill 2021—2022

Exploiting Task Affinity for Efficient Multitask Learning on Low-Resource Systems.
Advisor: Prof. Shahriar Nirjon

- Developed and evaluated an efficient multitask learning framework and memory management algorithm to optimize the orders of multitask executions on low-resource embedded systems.
- Paper accepted by *EWSN 2024*.

Undergraduate Research Assistant, Embedded Intelligence Lab, UNC-Chapel Hill 2021—2022

Efficient Neural Network Implementations on 16-bit Microcontrollers.
Advisor: Prof. Shahriar Nirjon

- Developed a neural network model generator for 16-bit TI MSP430 series microcontrollers. Improved workflow from minutes to several seconds.
- Reduced more than 50% inference time and energy consumption using hardware accelerator compared to the state-of-the-arts.
- Demo published in *IPSN 2022*.

Mentored Research, UNC-Chapel Hill 2021

Remote Collaborative Physics Simulation for High School Physics Education.
Advisor: Prof. Prasun Dewan

- Developed a remote user interface coupling platform for physics simulations in high school physics education using RPCs and IPCs to synchronize physical animations remotely.

Teaching & Mentoring

Research Mentoring, Hyperdimensional Computing on Embedded Systems Spring 2024
Mentee: Run Wang, undergraduate student in ECE at UCSD. Publication in *SenSys '24* poster.

Teaching Assistant, COMP 301: Foundations of Programming, UNC-Chapel Hill Summer 2021
Assisted in course instruction under Prof. Prasun Dewan, helping students in introductory-level Java programming.